Interview Guide (English)

Interviewer(s): _____	**Date: _____**
Interviewee(s): _____	**Location: _____**
Relation to project: _____	**Interview number: _____**
Stakeholder type: _____	**Language used: _____**

Thank you for taking the time to speak with us. We are a group of students from TU Delft, working together with UNAM Sisal, currently conducting research that supports in mapping mangrove change along the coast around Sisal. As part of this, we are speaking with different people who are involved with, or affected by, the mangroves in some way, to better understand how the mangroves, and the people connected to them, are changing over time.

Your answers will be used for our research and to help inform a tool we're building to share findings with researchers, local stakeholders, and policymakers. Participation is voluntary, and you're free to skip any question you'd rather not answer.

The interview will be conducted in Spanish. As we don't speak Spanish fluently ourselves, we'll be using translation support so we can each communicate in our own language. Because of this language barrier, the interview will likely follow more of a structured question-and-answer format, since we probably won't always be able to follow up on your answers.

General

1. Can you tell us a bit about yourself and your connection to the mangroves around Sisal?

Mangrove change and its drivers

2. In your opinion, how have the mangroves around Sisal changed over the past years or decades - in extent, density, or overall health?

3. What do you think has caused these changes?

   Separate human-caused factors (construction, land clearing, water pollution, tourism, aquaculture) from natural factors (storms, sea-level rise, drought).

4. Have you ever noticed a sudden or drastic change in the mangroves, rather than a gradual one - for example after a specific storm, construction project, or other event?
   - If yes: can you describe what happened and when?

5. Based on your own memory, could you pinpoint the areas within the mangroves that have changed the most? Draw them on a map and describe what, in your opinion, has caused these changes.

Communication

6. Have you had any communication or collaboration with other stakeholders regarding the mangroves or coastal development in Sisal? Think of policymakers, researchers, fishers, tour operators, developers, or NGOs.

7. (If the answer was affirmative) With whom, how frequently, and would you describe this communication as effective?

Present and future of the mangroves

8. Which of the following do you consider the biggest threat to the mangroves around Sisal? Rank from 'most' to 'least' concerning.
   - Construction and land-use change (housing, roads, infrastructure)
   - Water quality and pollution
   - Tourism pressure
   - Fishing and aquaculture practices
   - Natural hazards (storms, sea-level rise)
   - Illegal logging or resource extraction

STAKEHOLDER-SPECIFIC MODULES

Real Estate developers

A1. Can you tell us about any development or construction projects you have been involved with near or bordering the mangrove areas?

A2. Are you aware of any regulations or permitting requirements related to building near mangroves? How do these affect your projects?

A3. Have you ever had to adjust plans because of mangrove protection rules, or because of the condition of the mangroves themselves?

A4. How do you personally weigh economic development against mangrove conservation when making decisions?

Eco-tourism suppliers

B1. How central are the mangroves to the tours or services you offer? What would happen to your business if the mangroves continued to degrade?

B2. Have you noticed any changes in the mangroves that have affected your tours or routes (e.g. water levels, wildlife, accessibility)?

B3. What do you tell tourists about the mangrove ecosystem, and do you notice interest from tourists about its condition?

B4. Do you feel tourism activity itself puts pressure on the mangroves? If so, how do you try to manage that?

Fishers

C1. How dependent is your fishing on the mangrove areas (e.g. as nursery or breeding grounds)?

C2. Have you noticed changes in fish catch or species that you associate with changes in the mangroves?

C3. Are there any fishing or other local practices that you think affect the mangroves, positively or negatively?

Government & policymakers (e.g. municipality, environmental agency)

D1. What is the current policy or protected status of the mangrove areas around Sisal?

D2. What monitoring or data do you currently have on mangrove health, and how is it used in decision-making?

D3. What are the biggest challenges in enforcing protection or managing development near the mangroves?

D4. Would an early-warning tool for mangrove tipping points be useful for your policy work? What would make it usable in practice?

Local Residents

E1. How do you personally use the mangrove area (e.g. for wood, recreation, storm protection, fishing)?

E2. What changes have you personally noticed in the mangroves over the time you have lived here?

E3. What concerns, if any, do you have about the future of the mangroves for your community?

Researchers/ academics with prior work in this area (expert interview)

F1. Could you briefly describe the research you have conducted in or around Sisal's mangroves?

F2. What were your key findings on mangrove change, drivers of degradation, or resilience?

F3. Did your research identify any early-warning indicators or tipping points? What variables or thresholds did you use?

F4. What data or methods would you recommend we use - or avoid - based on your experience in this area?

F5. Would you be willing to share data or sources, or introduce us to others who might be relevant to our project?